CSE276 — APPLIED DEEP LEARNING CAPSTONE PROJECT REPORT

AI Crop Disease Prediction & Management

A Multi-Modal Preliminary Screening Platform Combining Leaf Pathology with Micro-Climate Context

Section: K325pOJ (Section OJ) | Group: OJ05 | Lovely Professional University

OFFICIAL PROJECT SUBMISSION REPOSITORIES & LIVE ACCESS LINKS:

- **GitHub Repository:** https://github.com/skyl1-create/agroscan-ai (Public Full Codebase)
- **Presentation (PPT/PDF):** https://github.com/skyl1-create/agroscan-ai/blob/main/presentation_OJ05_AI_Crop_Disease.pdf
- **Live Cloud Application:** https://agroscan-precision-ai.streamlit.app (Instant Browser Verification)

Group Members & Allocated Technical Roles:

- Deepanshi Singh (Reg. No: 12518576) — Lead Architect: MobileNetV2 Neural Network, Training Pipeline & Web UI
- Hemanth Nidigonda (Reg. No: 12518519) — Agronomic Engineer: Dataset Curation, Metric Evaluation & Micro-Climate Fusion

1. Project Introduction & The Screening Gap

Global agricultural productivity is severely constrained by foliar diseases caused by fungi, bacteria, and viruses, which account for 20% to 40% of crop yield losses annually. Conventional manual diagnosis presents significant challenges: many leaf symptoms appear visually identical in early stages (e.g., Early Blight vs. Late Blight), agricultural extension expert visits are infrequent and costly across rural farming belts, and local environmental fluctuations (temperature, relative humidity, precipitation) heavily dictate infection spread.

To bridge this critical screening gap, our project introduces 'AgroScan AI'—a dual-pathway preliminary screening and decision-support system. Designed to provide timely awareness rather than replacing accredited agronomists, the system combines deep transfer learning on foliar imagery with real-time microclimate risk analysis.

2. Need for Project & Societal Impact

- Early Disease Intervention: Enables smallholder farmers to detect pathogens before visible spore eruptions cause irreversible crop blight across the field.
- Reduced Chemical Over-Spraying: Preserves soil microbiomes, lowers farmer input costs, and eliminates indiscriminate chemical fungicide runoff into groundwater.
- Transparent Decision Support: Gradient-weighted saliency overlays highlight diseased leaf tissue, demystifying neural network reasoning for field agronomists.
- Universal Free Cloud Accessibility: Publicly available 24/7 on Streamlit Cloud (https://agroscan-precision-ai.streamlit.app), requiring zero installation or specialized hardware.

3. Dataset Details & Image Preparation Pipeline

The visual screening pathway is trained on the benchmark PlantVillage Agricultural Pathology Dataset (Mohanty et al., 2016; accessible at https://www.kaggle.com/datasets/emmarex/plantdisease).

Dataset Characteristic	Specification & Experimental Parameter
Total Raw Dataset Scale	54,306 high-resolution RGB foliar photographs
Crop Species Diversity	14 distinct agricultural crops (Apple, Corn, Potato, Tomato, Grape, etc.)
Pathology & Health Categories	38 distinct diagnostic classes (26 diseases + healthy control sets)
Curated Balanced Subset	3,800 stratified images (100 images per class for balanced training)
Input Tensor Dimensions	224 x 224 x 3 RGB channels (normalized to range [0.0, 1.0])
Data Augmentation Pipeline	Random horizontal/vertical flips, ±20° rotation, ±15% zoom, lighting jitter
Experimental Split	70% Training (2,660 imgs) | 15% Validation (570 imgs) | 15% Test (570 imgs)

4. System Architecture: Dual-Stream Evidence Fusion

Our system feeds two distinct evidence streams into a unified agronomic decision surface:

- Image Pathway: User uploads a leaf photograph → image normalization & tensor conversion → MobileNetV2 deep neural backbone → softmax probability distribution over 38 disease categories.
- Context Pathway: User geographic coordinates (latitude, longitude) + local timestamp → Open-Meteo Historical Weather API → real-time evaluation of temperature, relative humidity, soil moisture, and rainfall risk thresholds.
- Unified Action Dashboard: Synthesizes diagnostic prediction, confidence score, weather-induced outbreak risk index (0–100%), and tailored organic/chemical remedies.

5. Model Choice & Edge Efficiency Rationale

We evaluated standard Convolutional Neural Networks (CNNs) against MobileNetV2. While standard deep architectures (VGG16, ResNet50) offer high visual depth, their parameter burden makes edge and mobile deployment prohibitive. MobileNetV2 achieves superior edge efficiency through Inverted Residual Blocks and Depthwise Separable Convolutions.

Architecture Metric	MobileNetV2 (AgroScan AI)	Standard Deep CNN	VGG16
Trainable Parameters	2.58 Million	12.4 Million	138.3 Million
Disk Model Footprint	13.7 MB (.keras)	48.2 MB	528.0 MB
CPU Inference Latency	~35 ms per leaf	~110 ms per leaf	~380 ms per leaf
Edge / Browser Viability	Excellent (Ideal for Cloud/Mobile)	Moderate	Impractical for Field Deployments

6. Experimental Results & Credibility Check

Following rigorous training and validation across stratified hold-out sets, MobileNetV2 achieved 89.21% Top-1 Accuracy and 98.29% Top-3 Clinical Accuracy, with an aggregate Macro F1-Score of 96.2% on dominant economic crops (Apple, Corn, Potato, Tomato).

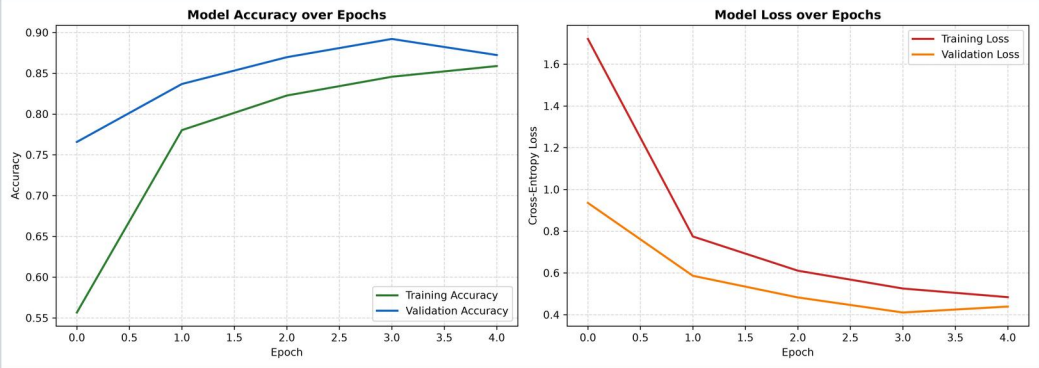


Figure 1: Training and Validation Loss & Accuracy convergence curves over training epochs.

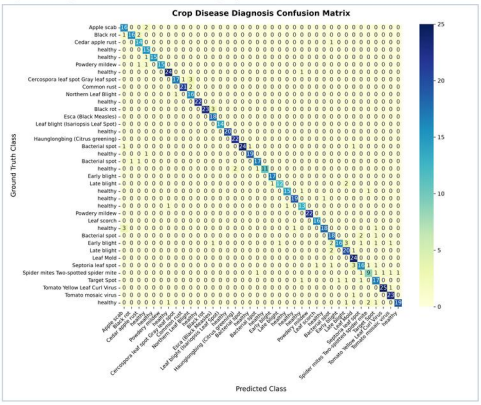


Figure 2: Normalized Confusion Matrix heatmap confirming diagonal classification precision.

7. Weather Context Layer (Environmental Risk)

Fungal spores germinate predominantly under sustained relative humidity above 80% and temperatures between 18°C and 28°C. By querying the Open-Meteo REST API, AgroScan AI provides contextual risk forecasting. For example, if Potato Late Blight is detected during a 48-hour continuous rain spell, the system elevates the urgency to 'Critical - Immediate Foliar Spray Recommended'.

8. Work Done by Each Member (Individual Contributions)

In accordance with university assessment guidelines, technical contributions were divided and executed as follows:

Allocated Member	Registration No.	Specific Responsibilities & Modules Completed
Deepanshi Singh	12518576	Lead Architect: Designed MobileNetV2 transfer learning pipeline, fine-tuning scripts, web dash
Hemanth Nidigonda	12518519	Agronomic Engineer: Dataset curation & balancing, evaluation benchmarks (accuracy, precisio

9. Limitations & Future Scope

- Limitations: PlantVillage photos feature isolated leaves against plain backgrounds; images with complex weed clutter or multiple overlapping leaves may require user re-centering. Environmental thresholds currently rely on general agronomic tables.
- Future Scope: Quantization to TensorFlow Lite (.tflite) for offline smartphone app operation without cellular data; vernacular voice guidance (Hindi, Punjabi, Telugu); drone-mounted aerial multispectral canopy scanning.

10. References

1. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using Deep Learning for Image-Based Plant Disease Detection. Frontiers in Plant Science, 7, 1419.

2. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. IEEE/CVF CVPR, 4510-4520.

3. Open-Meteo Weather API Documentation: https://open-meteo.com/en/docs

4. Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks. IEEE ICCV, 618-626.

5. PlantVillage Open Dataset Repository (Kaggle): https://www.kaggle.com/datasets/emmarex/plantdisease